Quran Echo

Project Team

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Chapter 1

Introduction

This document aims at outlining the technical and functional requirements for "Quran Echo", an AI-powered tool for enhancing the user's pronunciation and style of Quranic recitation through real time analysis and feedback.The primary readers of this document include:

- **Developers:** Tasked with implementing AI models for recitation style imitation, audio feature extraction, and ensuring precision in Makharij recognition.
- **Project Managers:** Responsible for managing timelines, resources, and ensuring the project progresses smoothly.
- **Marketing Staff:** Focused on positioning the product to appeal to users seeking to improve their recitation style and pronunciation.
- **Users:** Quran learners and enthusiasts who want to replicate the recitation style of famous reciters while focusing on accurate pronunciation.
- **Testers:** Ensuring the AI delivers precise feedback and the system functions as intended.
- **Documentation Writers:** Developing clear technical and user documentation to guide both developers and users.

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1.1 Problem Statement

In today's fast paced life, people rarely have time to invest in the proper learning of Quran and its recitation. Not everyone has access to teachers and the time it requires to improve one's Quranic recitation. There are some digital tools for Quranic learning focus primarily on recitation accuracy but often fail to provide comprehensive and precise feedback on correct pronunciation (Makharij) and teaching different recitation styles. This lack of detailed guidance limits learners' ability to achieve accurate and expressive recitation, resulting in difficulties in self-learning and maintaining motivation. Our project aims to address these gaps by developing an AI-powered solution that offers Makharij training, pronunciation correction and the ability to learn Qira'at styles, thereby enhancing the effectiveness and engagement of Quranic learning.

1.2 Scope

The Quran Echo project focuses on building an AI-powered Quranic recitation tool that enhances users' recitation skills through interactive feedback and progress tracking. The system will allow users to practice Quranic recitation, with specific emphasis on correcting pronunciation (Makharij) and applying Tajweed rules. Key features include real-time feedback for recitation, the ability to correct individual letter pronunciations (similar to the Qaidah method), and an imitation feature that lets users mimic the styles of famous Qaris. The tool will also support a Hifz module, enabling users to memorize Quranic verses and track their progress over time. Additional functionalities include error detection for common recitation mistakes, personalized guidance for improvement, and a user-friendly interface that ensures ease of learning and consistent practice. The project aims to make Quranic education more accessible by providing accurate feedback and progress visualization.

1.3 Modules

The proposed Quran Echo project consists of various modules, a summary is given below.

1.3.1 Client Web App

The Client Web App will be the primary interface for the users and it will provide the users with distinct functionalities and features. The app focuses on ease of use, offering seamless uploads and accessibility, making it more engaging.

1. User Login, Signup Profile Management
2. Recording and Uploading recitation
3. Real Time Recitation Feedback

4. Recitation Style Imitation
5. Makharij Recitation Correction

1.3.2 Admin Web App

The admin web app will be simple and robust, focusing on essential back-end tasks for minimal management purposes. The admin interface ensures that administrators can manage datasets, monitor basic user activities, and oversee system health without the need for extensive intervention.

1. Database Management
2. User Accounts Monitoring
3. System Performance Management

1.4 User Classes and Characteristics

Following are the user classes anticipated for the Quran Echo product, along with their pertinent characteristics:

User class	Description
Quran Learner	These users typically have limited knowledge of proper Tajweed and the correct pronunciation of Arabic letters (Makharij). They rely heavily on guided tools to help improve their recitation skills. For these users, ease of use is essential, and they benefit most from real-time feedback and personalized error correction. Many may also seek to imitate the recitation styles of renowned Qaris to enhance their fluency and confidence. This class of users is motivated by visible progress.. The Quran Echo platform serves as an accessible and interactive learning tool, making it easier for learners to refine their recitation skills at their own pace, whether they are just starting out or looking to improve..
Advanced Reciters	These users have significant experience with Quranic recitation and are well-versed in Tajweed rules and the correct pronunciation of Arabic letters (Makharij). They primarily use the platform for fine-tuning their recitation, focusing on details such as phoneme duration and melody. These users appreciate precise, technical feedback and are less reliant on general guided tools. Many seek to imitate renowned Qaris for perfection in tone, pitch, and rhythm. Their motivation comes from mastering the finer aspects of recitation, making the detailed feedback from Quran Echo essential for their advanced learning needs.
Hifz Practitioners	This group focuses on memorizing (Hifz) the Quran, often working on committing new sections to memory or revising previously memorized content. They rely on the Hifz module to monitor their progress and identify areas where further revision is required. These users benefit from features that allow them to track specific verses or chapters, receive reminders for revision, and visualize their memorization progress. Motivated by the goal of completing Hifz, they appreciate clear, structured feedback on their retention and performance. Quran Echo helps them stay organized and on track with their memorization goals.

Chapter 2

Project Requirements

This chapter describes the functional and non-functional requirements of the project.

2.1 Use-case Diagram

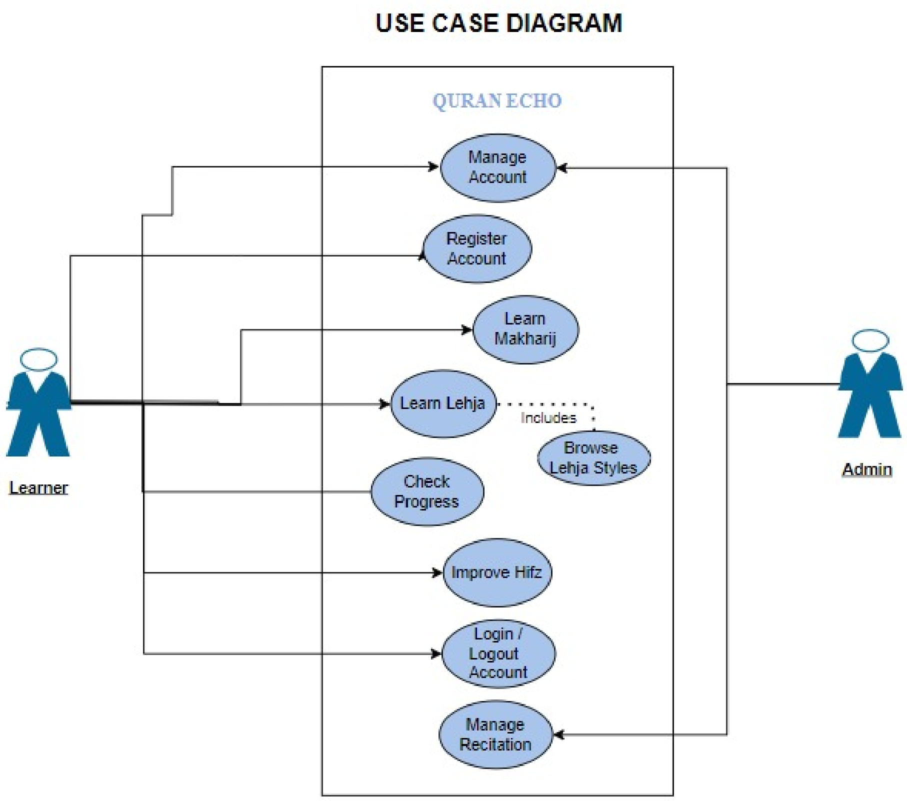


Figure 2.1: Use Case Diagram

2.2 Functional Requirements

This section describes the functional requirements of the *Quran Echo* system, organized by key features. Each module highlights specific functionalities that users will interact with.

2.2.1 Client Web App

2.2.1.1 Module 1: User Registration and Profile Management

This module manages user registration, login, and profile customization for personalized Quranic recitation practice.

1. FR1: The system shall allow users to create an account using their email or social media accounts.
2. FR2: The system shall enable users to log in, log out, and recover forgotten passwords.
3. FR3: Users shall be able to customize their profiles, including preferences for recitation tracking and Hifz progress.
4. FR4: The system shall store and display user recitation progress, including past uploads, Makharij corrections, and Hifz tracking.

2.2.1.2 Module 2: Makharij Correction for Individual Haroof

This module provides real-time feedback on the pronunciation of individual Haroof (letters) in a manner similar to the Qaidah method.

1. FR1: The system shall allow users to practice individual Haroof and receive real-time feedback on Makharij correction.
2. FR2: The system shall provide specific guidance on the correct articulation points (Makharij) for each letter.
3. FR3: Visual and audio feedback shall highlight mistakes in letter pronunciation and suggest corrections.
4. FR4: Users shall be able to repeat practice until the correct pronunciation is achieved, with progress tracking for individual Haroof.

2.2.1.3 Module 3: Recitation Recording and Upload

This module allows users to upload Quranic recitations for detailed analysis and feedback on Tajweed and Makharij.

1. FR1: The system shall allow users to upload recitation audio files in formats like MP3 and WAV.
2. FR2: The system shall analyze the uploaded audio and provide detailed feedback on errors and corrections.
3. FR3: Users shall receive a breakdown of Tajweed and pronunciation errors in a clear and structured format.

2.2.1.4 Module 4: Recitation Style Imitation

This module enables users to practice and imitate the recitation styles of famous Qaris, with feedback on style, pitch, and rhythm.

1. FR1: The system shall allow users to select a Qari's recitation style to imitate.
2. FR2: The system shall compare the user's recitation with the selected Qari's style, focusing on pitch, tone, and rhythm.
3. FR3: The system shall track progress and provide feedback on improvements over time.

2.2.1.5 Module 5: Hifz Module

This module supports users in memorizing (Hifz) the Quran and tracks their progress.

1. FR1: The system shall allow users to select specific verses or surahs for Hifz.
2. FR2: The system shall track the user's memorization progress, highlighting sections where improvement is needed.
3. FR3: Users shall receive reminders and suggestions for revising previously memorized sections.
4. FR4: The system shall display progress tracking for Hifz completion, showing how much has been memorized and retained.

2.3 Non-Functional Requirements

This section outlines the non-functional requirements of the Quran Echo project. These requirements ensure the system's quality and operational effectiveness. Each requirement is specific, measurable, and verifiable to guide the system's overall performance and reliability.

2.3.1 Reliability

The system should ensure high reliability, minimizing the frequency of failures during user interaction.

The Quran Echo platform will have an MTBF (Mean Time Between Failures) of at least 1,000 hours. In case of system failures, the user should be able to restart their session from the last saved progress. Failures include interruptions during audio processing or real-time feedback. The system will include error detection strategies, such as logging errors and alerting the development team for quick correction.

2.3.2 Usability

The system must be user-friendly, focusing on ease of learning and interaction to provide an efficient and engaging user experience.

The system shall allow a user to upload a recitation file or begin a live recitation feedback session within three clicks. The platform will include clear instructions, reducing the need for external tutorials, and providing on-screen error guidance. The interface will be designed for accessibility, including options for different text sizes and screen readers, ensuring inclusivity for users with disabilities.

2.3.3 Performance

The system's performance requirements focus on ensuring that user interactions with the platform are quick and responsive, providing feedback in real-time.

The system will provide real-time recitation feedback within few seconds after the user finishes a phrase. The audio file upload feature will support files up to 10MB and process them for feedback within 5 seconds. The user interface shall load fully in less than 3 seconds on a standard internet connection (5 Mbps).

2.3.4 Security

Security requirements ensure that user data, including recitation history and personal profiles, are well-protected.

The system shall prevent unauthorized access by implementing role-based access control (RBAC) to user accounts. All user data, including recitation audio, shall be encrypted during transmission and storage to prevent data breaches. The system will have measures in place to prevent unauthorized access, with an estimated time to breach of no less than 24 hours for a skilled attacker using standard tools.

2.4 Domain Model

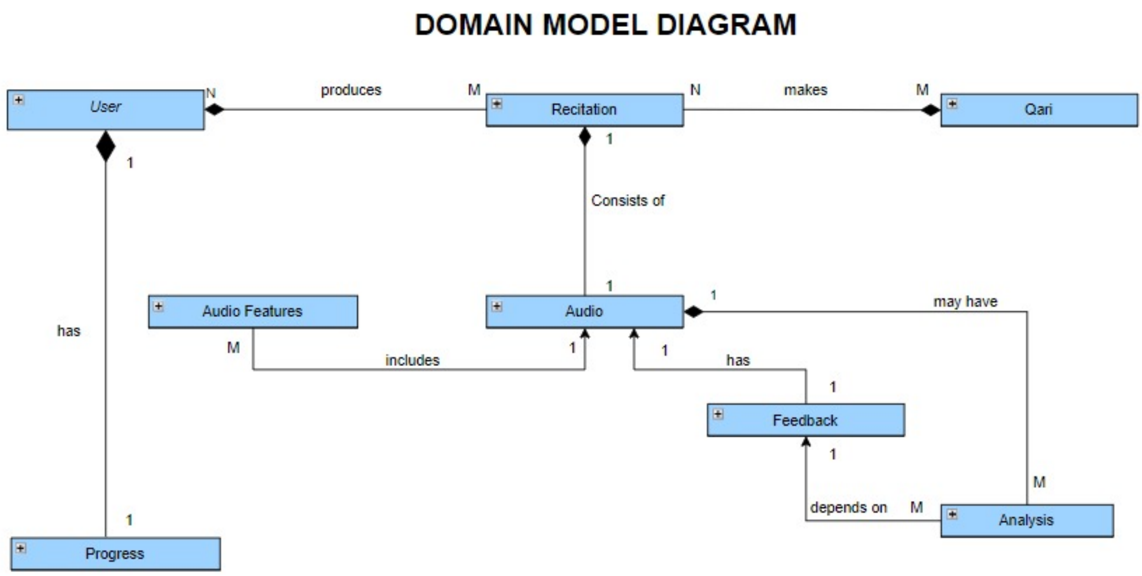


Figure 2.2: Domain Model

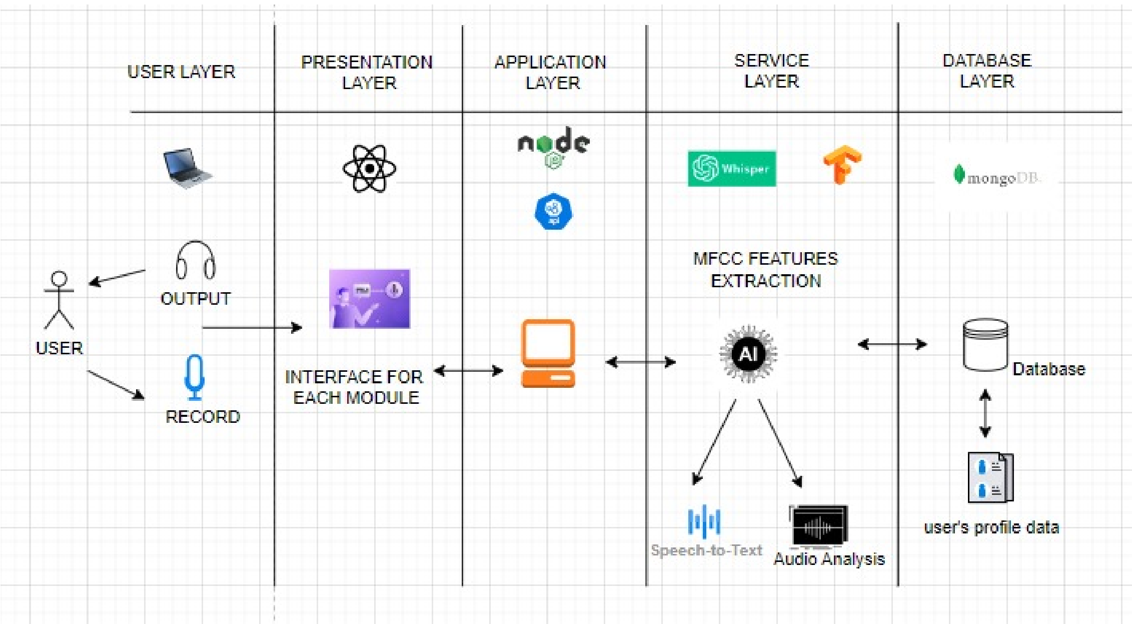
Chapter 3

System Overview

Provide any background information if necessary.

3.1 Architectural Design

The following diagram gives a detailed overview of the architectural design for this project.



3.2 Design Models

We will be using object oriented approach for this product.

- **Activity Diagram:** Illustrates the workflow of processes and activities within the system.
- **Class Diagram:** Represents the system's static structure, showcasing the classes, their attributes, methods, and the relationships between them.
- **Class-level Sequence Diagram:** Demonstrates the dynamic behavior of specific classes by detailing the interactions and method calls between objects in response to various events.

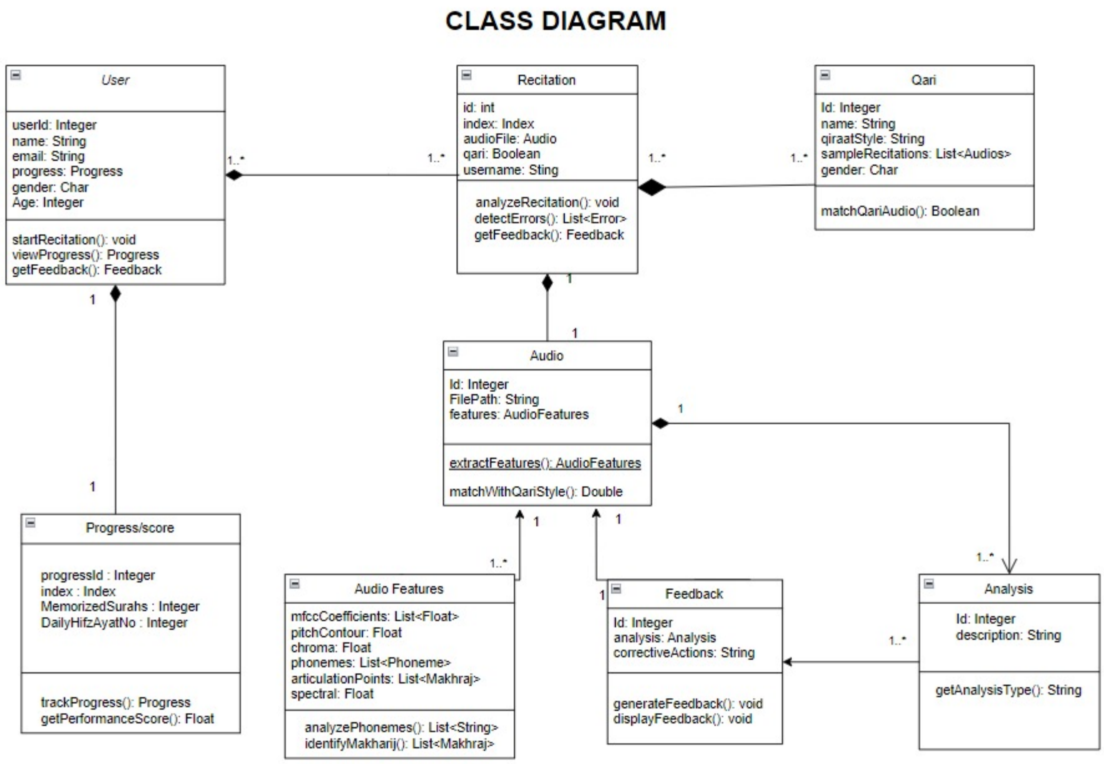


Figure 3.1: Class Diagram

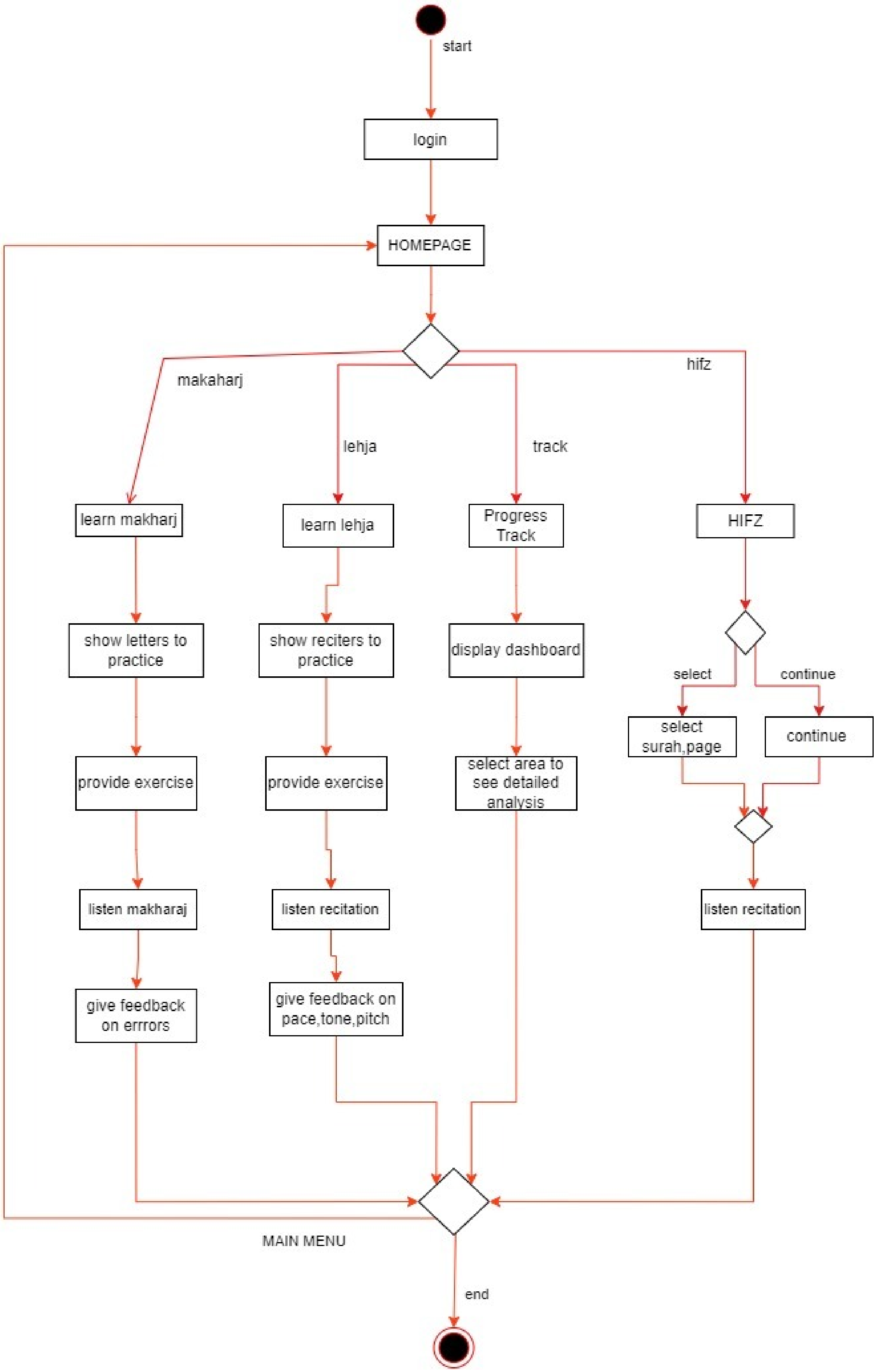


Figure 3.2: Activity Diagram

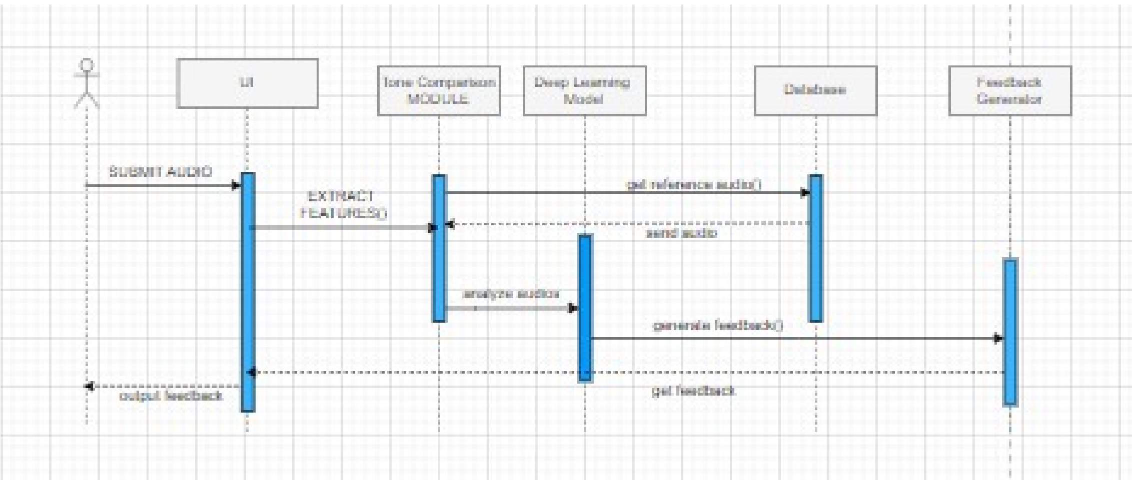


Figure 3.3: Sequence Diagram (Lehja Matching)

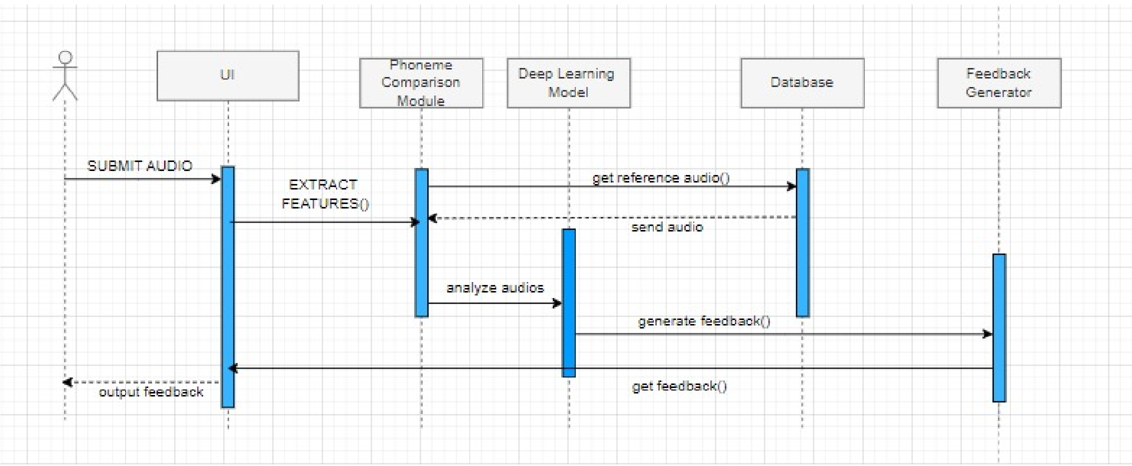


Figure 3.4: Sequence Diagram (Makharij)

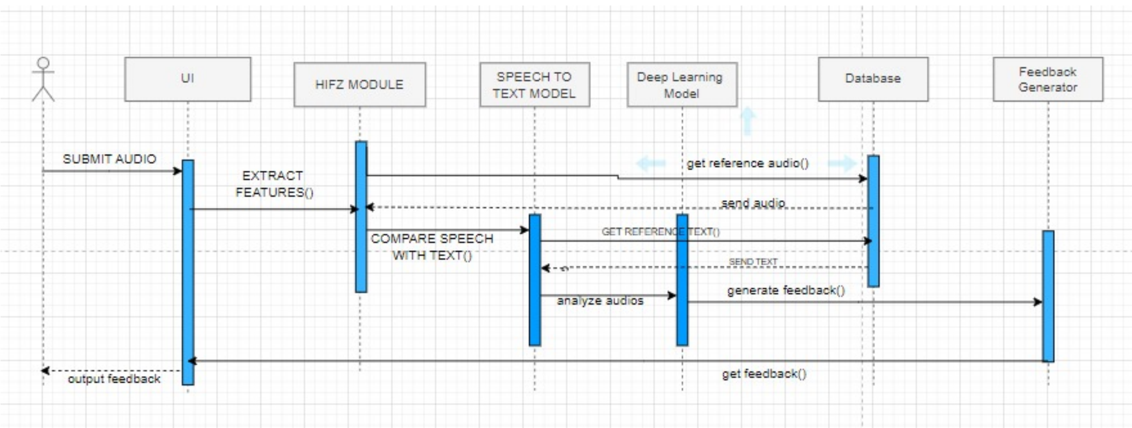


Figure 3.5: Sequence Diagram (Hifz)

3.3 Data Design

The information domain of the Quran Echo system is transformed into data structures that efficiently store and organize major data entities. Key entities include:

3.3.1 User Profiles

Stored as documents in a MongoDB collection, containing user credentials, preferences, and progress tracking.

3.3.2 Recitation sessions

Each session is a document referencing the user ID, session date, audio file path, and feedback entries related to pronunciation and Tajweed corrections.

3.3.3 Feedback Records

Feedback from recitation sessions is stored in a separate collection, detailing session ID, errors detected, and improvement suggestions.

3.3.4 Hifz Progress

Memorization progress is tracked in a dedicated collection, including user ID, selected verses, and memorization status.

The system will utilize MongoDB as its primary database, leveraging its flexible document-based storage for dynamic user data management and efficient retrieval, thereby enhancing the interactive learning experience.

Entity Relationship Diagram

[3]. [2] [1].

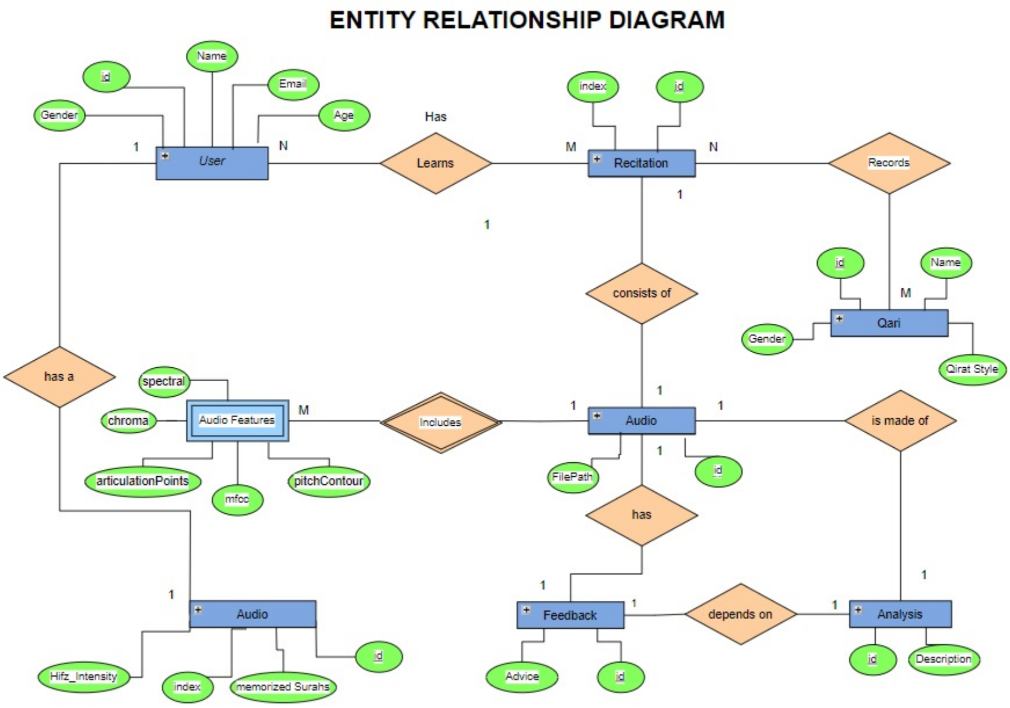


Figure 3.6: ERD

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[2] M. Al-Sharif and A. Al-Ghamdi. End-to-end quranic recitation recognition using cnn-bidirectional gru. *IEEE Access*.

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